GATE-AIII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 07



By- Rahul sir

Recap of previous lecture



Topic Continuity of a function

Topic Discontinuity of a function

Topics to be covered



Topic Mean value theorem + Differentiability

Topic Maxima and Minima of a function

Topic Problems based on mean value theorem

Topic Problems based on Maxima and minima

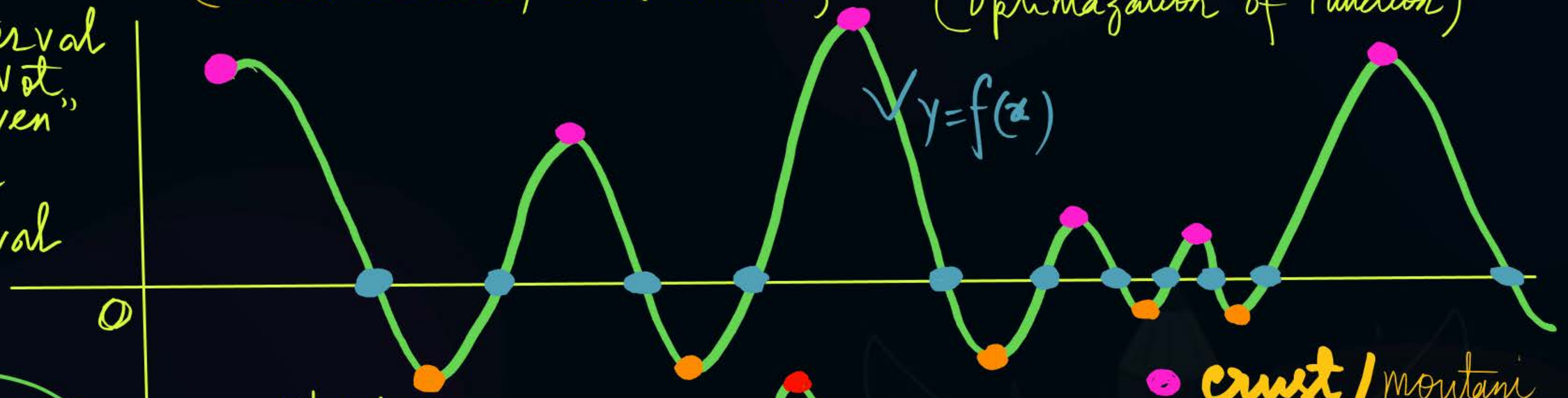
Max / Minima of a Function

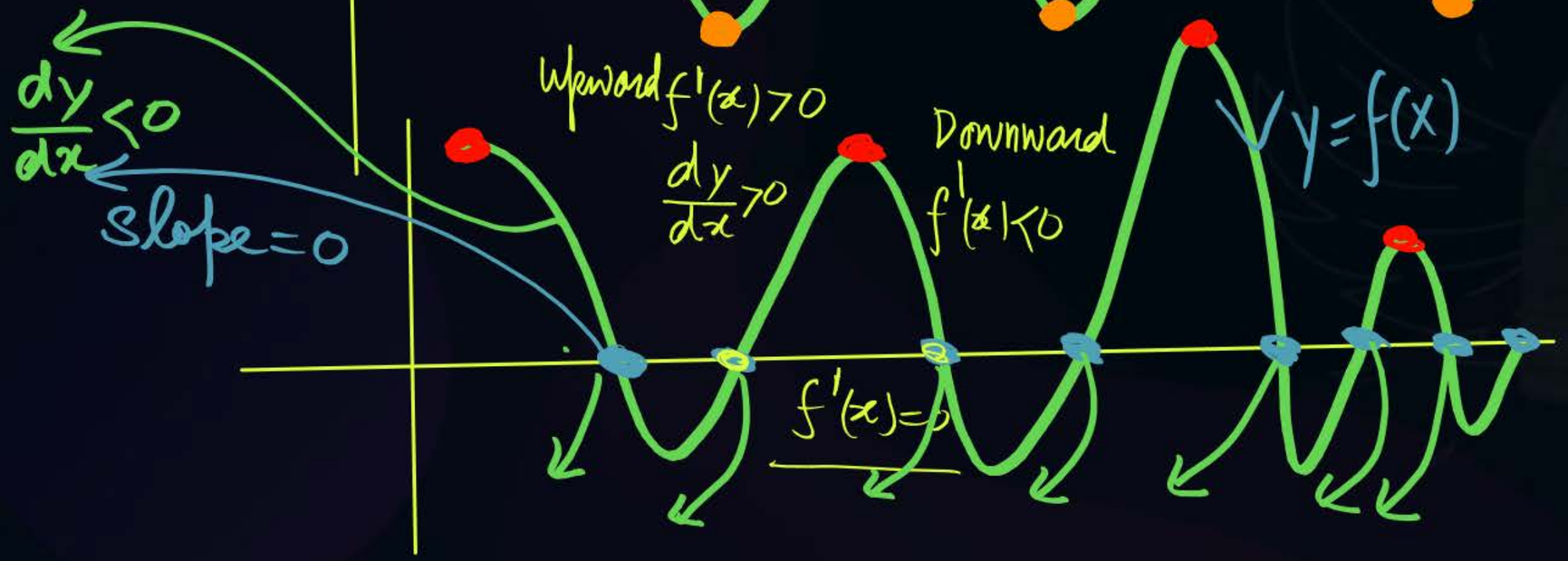
(Local maxima / Local minima)

Relative max or Relative min
(Optimazation of Function)

"Interval is Not given"

"Open Interval

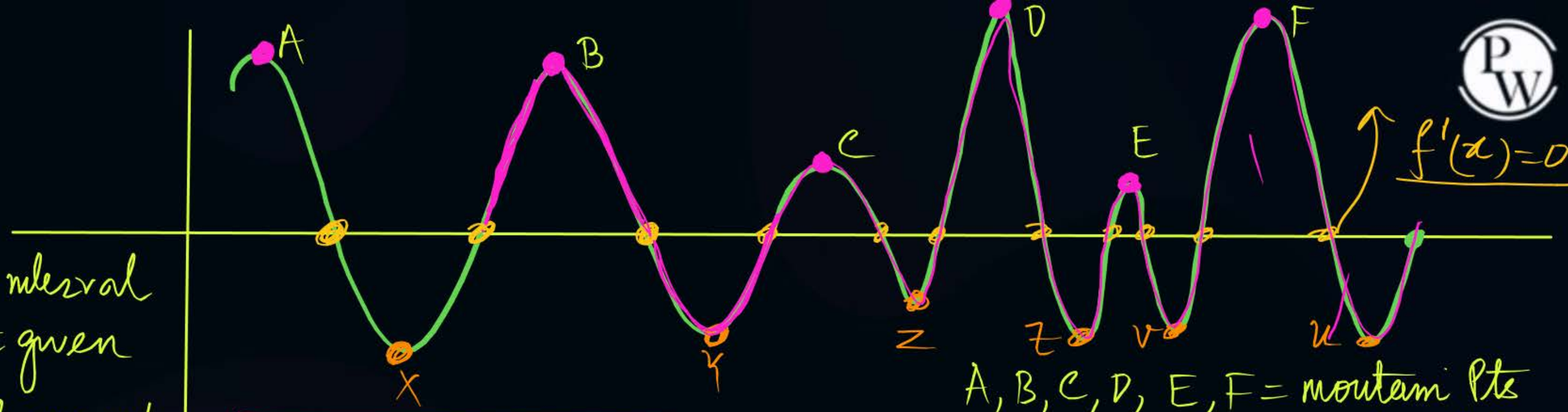




- crest / mountain
- Trough / valley
- stationary Pt or Critical pts

$f'(x) > 0$ (steadily increasing)

$f'(x) < 0$ (decreasing)



If nterval is Not given
= Local max / Local min
OR
Open Interval

Local max / Local min
Relative max / Local min

A, B, C, D, E, F = mountain Pts
= max (local)

X, Y, Z, T, U, V = minima (local)

$f'(a) = 0$ ⟨ Nature Change

$+ \longrightarrow -$

$- \longrightarrow +$

$\frac{dy}{dx} \rightarrow -\left(\frac{dy}{dx}\right)$

$\frac{dy}{dx} \leftarrow$

How to Evaluate The Local max/Local min (Pts of Extrema)

Step (A) If given Function $y = f(x)$

$f'(x) = 0$ to get The stationary Pts $\rightarrow x_0$

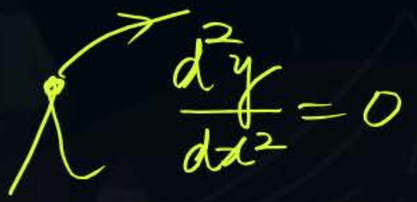
Step (B)

$f''(x_0) > 0$ (Minima)

$f''(x_0) < 0$ (Maxima)

$f''(x_0) = 0$ (Pts of Inflection)

(Neither max Nor minima)



Pt of Contraflexure

BMD Diagram ene

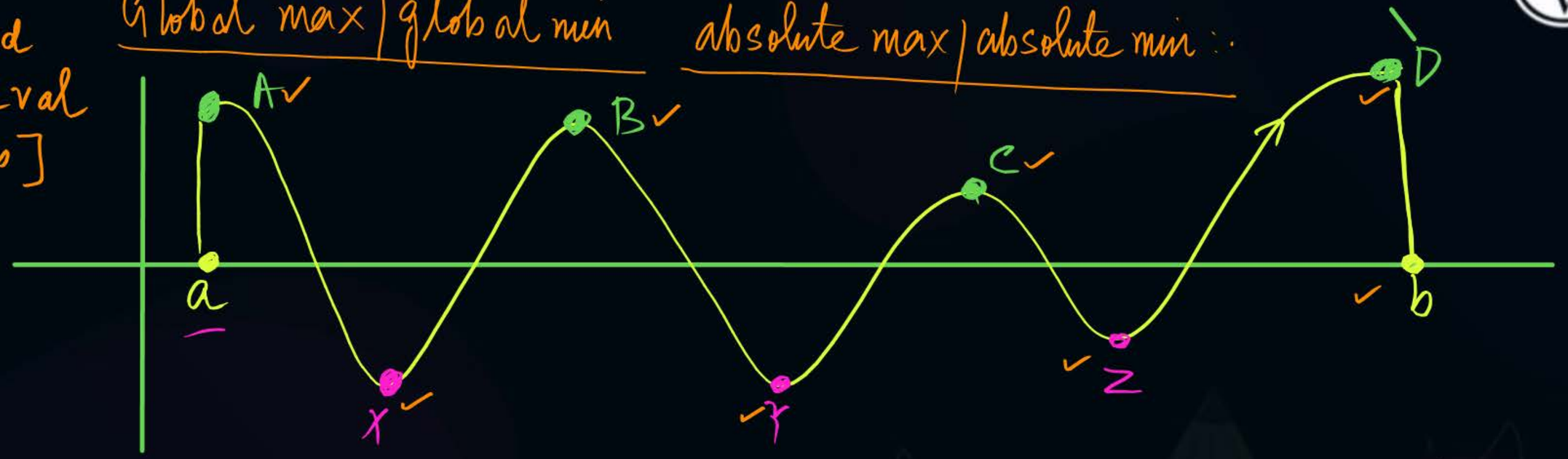
Graph-change mechanis

Step ③ max value $y = f(x_0)$

min value $y = f(x_0)$

Closed Interval $[a, b]$ = global

Global max / global min absolute max / absolute min :



Step ① If given Function $y = f(x)$

$$\frac{dy}{dx} = 0 \text{ (stationary Pt)}$$

Step ② global max = [max value of $f(A), f(B), f(C), f(D)$]

global min = [min value $f(x), f(Y), f(Z)$]

Absolute max → global max

Absolute min → global min

Topic : Single Variable Calculus

"Interval is Not given"

#Q. For the function $f(x) = x^2e^{-x}$, the maximum occurs when x is equal to

Local min / max

$f(x) = x^2e^{-x}$

Pts of Extrema

$y = x^2e^{-x}$ (I II)

STEP ① Diff. w.r.t to x

$f'(x) = x^2\frac{d}{dx}(e^{-x}) + e^{-x}\frac{d}{dx}x^2$

$y = \frac{d}{dx}(I \cdot II)$

$= I\frac{d}{dx}(II) + (II)\frac{d}{dx}(I)$

$\boxed{f'(x) = -x^2e^{-x} + e^{-x} \cdot 2x}$

$-x^2e^{-x} + e^{-x}2x = 0$

$e^{-x}[2x - x^2] = 0$

$2x - x^2 = 0 \quad x = 0$

$x(2-x) = 0 \quad x = 2$

Stationary Points

A 2

B 1

C 0

D -1

$$f'(x) = -x^2e^{-x} + 2xe^{-x}$$

$$f''(x) = -\frac{d}{dx}\underset{\text{I II}}{(x^2e^{-x})} + \frac{d}{dx}\underset{\text{I II}}{(2xe^{-x})}$$

$$= -\left[-x^2e^{-x} + 2xe^{-x}\right) + 2\left[1\,e^{-x} + (-e^{-x})x\right]$$

$$= x^2e^{-x} - 2xe^{-x} + 2e^{-x} - 2xe^{-x}$$

$$f''(x) = x^2e^{-x} - 4xe^{-x} + 2e^{-x}$$

at Pt $x=2$

$$f''(2) = 4e^{-2} - 8e^{-2} + 2e^{-2}$$

$$= -2e^{-2}$$

$f''(2) < 0$ (max at Pt $x=2$)

Max value $y = f(x) = x^2e^{-x}$

$$f(2) = 4e^{-2} = \frac{4}{e^2}$$

$f''(x) = 2 > 0$

minima

$f''(0) = 2 < 0$

$y = 0$



Topic : Single Variable Calculus

Closed interval
global max/min

#Q. The maximum value of $f(x) = x^3 - 9x^2 + 24x + 5$ in the interval $[1, 6]$ is

A 21

B 25

C 41

D 46

$f(x) = x^3 - 9x^2 + 24x + 5$ In the Interval $[1, 6]$

Step: 01

$f'(x) = 0$

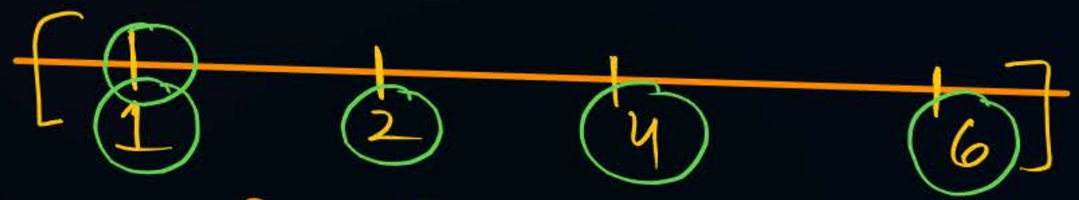
$3x^2 - 18x + 24 = 0$

$3(x^2 - 6x + 8) = 0$

$\Rightarrow x^2 - 6x + 8 = 0$

$(x-2)(x-4) = 0$

$\boxed{x = 2, 4}$



$f(x) = x^3 - 9x^2 + 24x + 5 =$

$f(1) = 1^3 - 9\times1^2 + 24\times1 + 5 = 21$

$f(2) = 2^3 - 9\times2^2 + 24\times2 + 5 = 25$

$f(4) = 4^3 - 9\times4^2 + 24\times4 + 5 = 21$

$f(6) = 6^3 - 9\times6^3 + 24\times6 + 5 = 41$

Max value

Maximum Value = at $x=6$ = (41)

min value at $x=1, 4$ = (21)



Topic : Single Variable Calculus

#Q. For $0 \le t < \infty$, the maximum value of the function $f(t) = e^{-t} - 2e^{-2t}$ occurs at

A $t = \log_e 4$

B $t = \log_e 2$

C $t = 0$

D $t = \log_e 8$

$f(t) = e^{-t} - 2e^{-2t}$

$f'(t) = -e^{-t} + 4e^{-2t} = 0$

$-e^{-t} + 4e^{-2t} = 0$

$\Rightarrow e^{-t}[-1 + 4e^{-t}] = 0$

$-1 + 4e^{-t} = 0$

$4e^{-t} = 1$

$e^{-t} = \frac{1}{4}$

$e^{t} = 4$

$t = \log_e 4$

$f'(t) = -e^{-t} + 4e^{-2t} = 0$

$t = \log_e 4$

$f''(t) < 0$

max

$t = \log_e 4$



Topic : Single Variable Calculus

#Q. The maximum value of the function $f(x) = \ln(1 + x) - x$ (where $x > -$

1) occurs at x = ____.

Max value
$= f(x)$
$= \ln(1+x) - x$
$= \ln 1$
$f(\max) = 0$

$f(x) = \ln(1+x) - x$

$f'(x) = \frac{1}{(1+x)} - 1 = 0$

$\frac{1}{1+x} = 1$

$1 = 1+x$

$x+1-1 = 0$

$\boxed{x = 0}$ stationary Pt

Local

$f''(x) = \frac{-1}{(1+x)^2}$

$f''(0) = -1 < 0$

max $x = 0$



Topic : Single Variable Calculus

global

#Q. The maximum value of $f(x) = 2x^3 - 9x^2 + 12x - 3$ in the interval $0 \le x \le 3$ is ____.

$f(x) = 2x^3 - 9x^2 + 12x - 3$

$f'(x) = 6x^2 - 18x + 12$

$= 6(x^2 - 3x + 2) = 0$

$x^2 - 3x + 2 = 0$

$x^2 - 2x - x + 2 = 0$

$x(x-2) - 1(x-2) = 0$

$x = 1$

$x = 2$



$f(0) = 2(0)^3 - 9(0)^2 + 12 \times 0 - 3 = -3$

$f(1) = 2 \times 1^3 - 9 \times 1^2 + 12 \times 1 - 3 = 2$

$f(2) = 2 \times 2^3 - 9 \times 2^2 + 12 \times 2 - 3 = 1$

$f(3) = 6$

$-3, 2, 1, 6$

max value $x = 3 \xrightarrow{\max} 6$



Topic : Single Variable Calculus

$x = 1$

#Q. Let $f(x) = xe^{-x}$. The maximum value of the function in the interval $(0, \infty)$ is

$f(x) = xe^{-x}$

$(0, \infty)$ open interval

$f'(x) = 0$

Local max/min

$-xe^{-x} + e^{-x} = 0$

Max value $= f(1) = 1e^{-1} = \frac{1}{e}$

$e^{-x}(-x+1) = 0$

$\checkmark \boxed{x = 1}$

$f''(x) = -(xe^{-x} + e^{-x}) - e^{-x}$

$f''(1) = xe^{-x} - 2e^{-x}$

$= e^{-1} - 2e^{-1} = \frac{-1}{e} < 0$ Max

A e^{-1}

B e

C $1 - e^{-1}$

D $1 + e^{-1}$

Topic : Single Variable Calculus

4 to 6

✓ Jain & Iyenger

✓ Erwin Kreszig | micheal greenberg

#Q. The maximum value attained by the function $f(x) = x(x-1)(x-2)$ in the interval [1,2] is.

$f(x) = (x^2 - x)(x-2)$

$= x^3 - 2x^2 - x^2 + 2x$

$f(x) = x^3 - 3x^2 + 2x$ [1, 2]

$f'(x) = 3x^2 - 6x + 2$ global max

$3x^2 - 6x + 2 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{6 \pm \sqrt{36 - 4 \times 3 \times 2}}{6}$

$x = \frac{6 \pm \sqrt{36 - 24}}{6}$

$x = \frac{6 \pm \sqrt{12}}{6}$ Roots

$\frac{2}{6_3}$

$x = \frac{6 \pm 2\sqrt{3}}{6}$

$= \frac{6\left(1 \pm \frac{1}{\sqrt{3}}\right)}{6}$

$x = 1 \pm \frac{1}{\sqrt{3}}$

global 1, $1+\frac{1}{\sqrt{3}}$, 2



Topic : Single Variable Calculus

Gate

H.W

#Q. The minimum value of the function $f(x)=\frac{1}{3}x\left(x^2-3\right)$ in the interval $-100 \le x \le 1000$ occurs at x = _____.



Topic : Single Variable Calculus

global

H.W

#Q. Let $f(x) = 3x^3 - 7x^2 + 5x + 6$. The maximum value of $f(x)$ over the interval $[0,2]$ is _____. (upto 1 decimal place)



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

✓ Max/min
- Local (open)
- global (closed)

✓ Problems.

THANK - YOU